

Robust Federated Chest-Radiograph Learning Under Institutional Heterogeneity: Optimization, Calibration, and Client-Level Reliability

Prospective MSc Thesis Proposal

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Research area: trustworthy and privacy-preserving machine learning

Proposal Summary

Federated learning allows institutions to train a shared model without pooling their raw data, but its performance can become unstable when participating clients have different data distributions. In medical imaging, these differences may reflect disease prevalence, patient populations, imaging equipment, acquisition protocol, view position, or clinical workflow.

My completed research project provides a reproducible starting point for studying this problem. It showed that stronger label heterogeneity reduced chest-radiograph performance and that FedProx reduced client drift by approximately one-half without consistently improving PR-AUC. I propose to investigate when optimization stability becomes practically useful. The study will compare FedAvg and FedProx across several proximal strengths and local-training budgets, introduce controlled label and view-position heterogeneity, and evaluate calibration, threshold transfer, worst-client performance, and external generalization where feasible.

1. Background and Rationale

Federated learning supports collaborative model training without requiring each institution to transfer raw data to a central repository [1], [3]-[5]. It is attractive for medical imaging because data sharing is constrained by privacy, governance, ownership, and institutional policy. However, keeping data local does not remove statistical heterogeneity. When clients optimize different local distributions, their updates may diverge and the global model may perform unevenly across institutions.

FedProx was introduced to reduce this divergence by penalizing large departures from the current global model [2]. My completed project confirmed that the method can substantially reduce update magnitudes. It also showed that lower model movement is not automatically equivalent to better generalization. This creates a focused research problem: under which conditions does stronger optimization control improve discrimination, calibration, threshold transfer, and client-level reliability?

2. Preliminary Work and Motivation

As preparation for the proposed research, I completed a patient-aware study using the RSNA Pneumonia Detection Challenge dataset. The project included:

- 26,684 chest-radiograph examinations linked to 11,452 original patients;
- patient-exclusive training, validation, and test splits with zero overlap;
- five-client IID, moderate non-IID, and severe non-IID partitions;
- centralized, FedAvg, and FedProx experiments across three random seeds;
- validation-only checkpoint and threshold selection; and
- one frozen final evaluation of 18 selected models.

Condition	ROC-AUC	PR-AUC	Balanced accuracy	F1-score
Centralized	0.8295 ± 0.0040	0.5800 ± 0.0057	0.7510 ± 0.0037	0.5745 ± 0.0107
FedAvg IID	0.8179 ± 0.0030	0.5605 ± 0.0042	0.7453 ± 0.0048	0.5695 ± 0.0068
FedAvg moderate	0.8131 ± 0.0027	0.5553 ± 0.0065	0.7355 ± 0.0032	0.5541 ± 0.0055
FedAvg severe	0.8070 ± 0.0005	0.5440 ± 0.0043	0.7312 ± 0.0020	0.5529 ± 0.0029
FedProx moderate	0.8112 ± 0.0020	0.5505 ± 0.0040	0.7375 ± 0.0050	0.5608 ± 0.0036
FedProx severe	0.8076 ± 0.0019	0.5428 ± 0.0092	0.7335 ± 0.0030	0.5581 ± 0.0076

The results showed that FedProx improved optimization stability and achieved slightly higher balanced accuracy and F1-score, while FedAvg retained a small PR-AUC advantage. Neither method consistently dominated across all predictive metrics. This mixed result motivates a deeper investigation of when reduced client drift produces practically meaningful improvements rather than only smaller model updates.

3. Research Gap and Novelty

Many federated-learning comparisons emphasize global discrimination metrics while giving less attention to calibration, threshold transfer, or the weakest client. A model can achieve a reasonable global AUC yet remain unreliable for a small or unusual institution. The proposed study will jointly examine four connected dimensions: client heterogeneity, optimization stability, global discrimination, and client-level reliability.

Proposed novelty: the study will test whether reduced update magnitude is associated with better calibration, threshold transfer, and worst-client performance across controlled forms of medical-imaging heterogeneity. This will move the analysis beyond the question of which algorithm has the highest average AUC.

4. Aim

To develop and evaluate a reproducible federated chest-radiograph framework that explains how institutional heterogeneity and optimization choices affect discrimination, calibration, threshold transfer, and client-level reliability.

5. Objectives

1. Extend the verified patient-aware baseline without reusing the frozen test set for model development.
2. Compare FedAvg and FedProx across prespecified proximal coefficients and local-epoch settings.
3. Model both label skew and AP/PA view-position skew.

4. Measure global performance, calibration, threshold transfer, and worst-client behavior.
5. Test whether client and global update magnitudes are associated with predictive generalization.
6. Conduct external validation where compatible data, licensing, and supervision permit.

6. Research Questions

1. How do label skew and AP/PA view-position skew affect global and client-level performance?
2. How sensitive is FedProx to the proximal coefficient and local-training duration?
3. When does lower client drift correspond to better PR-AUC, calibration, threshold transfer, or worst-client performance?
4. How well do validation-selected thresholds transfer across heterogeneous clients and, where feasible, external data?

7. Proposed Methodology

Data

The RSNA dataset will remain the primary resource because the official RSNA-to-NIH mapping supports patient-aware reconstruction. A second public chest-radiograph dataset may be used for external evaluation after checking label compatibility, data access, licensing, and supervisor approval. External validation will be treated as an extension rather than a requirement for the core study.

Heterogeneity scenarios

The experiments will include approximately IID clients, Dirichlet label skew, AP/PA view-position skew, and a combined scenario where computationally practical. All examinations from the same patient will remain within one split and one federated client.

Algorithms and configurations

FedAvg and FedProx will be compared using prespecified FedProx coefficients of 0.001, 0.01, and 0.1 and local-training budgets of one and three epochs. Configuration files, client partitions, and random seeds will be fixed before final evaluation. Additional methods may be considered only after the primary comparison is complete and with supervisor guidance.

Outcomes

Primary outcomes will include PR-AUC, ROC-AUC, balanced accuracy, F1-score, log loss, calibration error, threshold-transfer performance, worst-client performance, and client/global update magnitudes. Results will be reported across multiple random seeds, with descriptive uncertainty and careful limits on statistical claims.

Evaluation and reproducibility

Model development and threshold selection will use validation data only. Final configurations will be frozen before held-out evaluation. The repository will preserve configuration files, partition manifests, model hashes, aggregate tables, and figure-generation code. The existing frozen test set

will not be repeatedly reused for development; a revised split or external dataset will be considered if a new final evaluation is required.

8. Expected Contributions

- A patient-aware framework separating label-related and view-related heterogeneity.
- A practical analysis of FedProx strength and local-training duration.
- Evidence connecting update stability with calibration, threshold transfer, and client-level reliability.
- A transparent evaluation protocol that limits test-set reuse and preserves reproducibility.
- A foundation for later security and privacy extensions such as secure aggregation, differential privacy, or threat-aware federated learning.

9. Preparation and Research Fit

Through the completed project, I have developed practical experience in patient-aware medical-image preparation, federated optimization, controlled non-IID partitioning, multi-seed evaluation, validation-based model selection, experiment tracking, and scientific interpretation. This preparation positions me to undertake the proposed work under academic supervision and to extend it through stronger mathematical analysis, broader experiments, calibration assessment, and external validation.

The proposal is focused enough for an MSc thesis while also fitting my broader interest in secure and trustworthy intelligent systems. In the longer term, I am interested in AI and machine learning, cybersecurity, applied cryptography, privacy-preserving computation, and secure distributed systems—particularly where security and privacy mechanisms can improve the reliability of intelligent systems.

10. Ethics and Responsible Use

The research will use public, de-identified datasets and will not collect new patient information. The resulting models will remain research prototypes and will not be presented as clinical diagnostic systems. Institutional ethics, data-access, security, and responsible-use requirements will be confirmed with the supervisor before formal thesis work begins.

11. Indicative 12-Month Plan

Months	Activities
1-2	Literature review, supervisor refinement, ethics and access checks.
3-4	Implement label-skew and view-skew scenarios.
5-6	Run FedAvg/FedProx coefficient and local-epoch experiments.
7-8	Calibration, threshold-transfer, and worst-client analysis.
9	External validation where feasible.

10	Consolidate the reproducibility package.
11	Write the thesis and manuscript.
12	Revision, defense preparation, and submission.

References

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